

Ethan Carter Edwards

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Education

A.B. in Honors Computer Science, Harvard University (August 2025 - May 2029)

- 3.8 GPA; Coursework: Grad. Distributed Systems, Grad. Systems Security, Operating Systems, Intro Systems.

A.S. in Science, Wytheville Community College (June 2022 - May 2025)

- Completed concurrently in high school. 2025 John V. Neff Humanities Award. Phi Theta Kappa Honors.
- 4.0 GPA; Calculus, Chemistry, Cybersecurity, British/American Literature, Political Science, Art History

Advanced Diploma, Carroll County High School (August 2021 - May 2025)

- 4.7/4 GPA; Valedictorian; Programming, Physics, Math, Chemistry, Economics, Drafting
- Varsity Cross Country, Swim, Track, Competitive Math, and Scholastic Bowl Teams
- FBLA Nationals: Computer Problem Solving (3rd), Networking Infrastructures (5th), and Network Design (8th)
- FBLA Regional Vice President, Chapter President; SkillsUSA Web Development District Champion

Experience

Teaching Fellow for CS 61 Systems Programming, Harvard University (August 2026 - December 2026)

Infrastructure Software Engineer, Exa.ai (May 2026 - Current)

- Architected a monorepo build system to manage complex acyclic dependencies between 500+ projects with Nix.
- Designed and implemented a Kubernetes plugin to replace OCI images with Nix manifests, accelerating deployments 5x.
- Sped up build and CI times by 10-100x through leveraging and writing custom multi-derivation Nix builders and tooling.
- Implemented an AWS S3-backed caching server for Nix builds and Nix flake inputs using Go, Cloudflare, and k8s.
- Gave two presentations at DEFCON Nix Vegas demonstrating how we use Nix to model and manage our infrastructure.

Software Development Intern, Inigo (January 2026 - May 2026)

- Used Python, Flask, Postgres, Typescript, React, and Tanstack Router to develop a finance/invoicing platform.
- Improved codebase documentation, test coverage for faster development cycles.

Summer of Nix Fellow, NixOS Foundation (June 2025 - September 2025)

- Used Nix, NixOS, Rust, Python, and OCaml to support 50+ programs and modules for clients across the E.U.

IT Support Intern Team Lead, Carroll County Public Schools (March 2023 - August 2025)

- Managed a team of 6 interns across three buildings over the span of three years. Hired interns each year.
- Wrote PowerShell scripts to simplify software delivery process for district, saving hundreds of man-hours.
- Deployed 1,200+ Chromebooks and 80+ AV systems; conducted professional development training to faculty and staff.
- Livestreamed, recorded, and managed AV for graduations, board meetings, talent shows, and more.

Contributing Technical Writer, StorageReview.com (June 2021 - December 2021)

- Wrote 7 articles on networking equipment, enterprise and consumer storage options, and Homelab technologies.
- Benchmarked, tested, and reviewed enterprise and consumer server hosting software.

Systems Administrator Intern, AlthaTechnology (March 2020 - August 2021)

- Designed and implemented a distributed storage system serving up to 30 servers with 10Gb/s fiber network.
- Implemented distributed and highly available compute layer using Docker Swarm, reducing costs by \$300/month.

Projects and Activities

NixOS Linux/Nixpkgs Maintainer, Nix, Python, Rust, Open-Source

- Serve on Nix MacOS, CUDA, NGI, Tmux teams. Maintain 500+ packages. Reviewe 2500+ PRs. 450+ PRs merged.

Homelab, Home Networking, and Self-Hosting, Docker, ZFS, NFS, Proxmox, Networking

- Maintain a multi-architecture web-hosting fleet to deploy sites with 2K+ daily active users at a 99% uptime.
- Hardware includes Dell Poweredge R320, Cisco 3750G L3 Switch, and managed WiFi Access Points.
- Serve on Nix MacOS, CUDA, NGI, Tmux teams. Maintain 500+ packages. Reviewe 2500+ PRs. 450+ PRs merged.

The Ridge Podcast, NextJS, React, Flask, Cloudflare R2, SCSS, Docker, REST APIs

[Website](#)

- Developed a custom podcast CMS to serve 500+ daily listeners with 99% uptime using a global CDN system.
- Interviewed 100+ politicians, historians, veterans, musicians, journalists, writers, teachers, and more.

The Harvard Crimson Tech Board, Django, React, Cloudflare R2, GraphQL

[Website](#)

- Migrated site serving 21 million annual viewers from Amazon S3 to Cloudflare R2, saving \$400/month.
- Debugged cache invalidation issues, developed interactive article editor, ArchieML parser.

Dracula Tmux, Bash, REST APIs, Open-Source

[850+ GitHub Stars](#)

- Maintainer for a popular Open-Source modular tmux plugin for Linux and *BSD systems.
- Developed statuses for Kubernetes, Git, Terraform, Weather, GPU/RAM/CPU utilization, and Blood Glucose levels.

HotBillsNearYou.com, YHack 2026 winner, NextJS, React, Flask, Supabase

[Website](#)

- “Tinder For Legislation”. Created an Agentic AI platform that summarizes and displays legislation to vote on.

Traxos, Multi-Paxos Distributed BitTorrent Tracker

- Fully-featured BitTorrent Tracker written in C++ using the Multi-Paxos Distributed Consensus Algorithm.

Certifications

- [FCC Amateur Radio Amateur Extra Class License](#)
- [CompTIA A+ \(1101-1102\)](#)
- [CompTIA Security+](#)
- [\(ISC\)² \(CC\)](#)

Talks/Panels/Presentations

1. Edwards, Ethan Carter “Trusting Trust and Security”. Harvard Systems Reading Group, Cambridge, MA, September 12, 2026.
2. Edwards, Ethan Carter “Your Infrastructure Is a DAG: Manage It With Nix”. DEF CON 34 Nix Vegas, Las Vegas, Nevada, August 8, 2026.
3. Edwards, Ethan Carter “Github Rate Limits Got You Down? FOD’s Failing to Build? Let’s Talk About Caching”. DEF CON 34 Nix Vegas, Las Vegas, Nevada, August 7, 2026.
4. Edwards, Ethan Carter “NixOS, Nixpkgs, and Nix”. Harvard Systems Reading Group, Cambridge, MA, February 21, 2026.
5. Edwards, Ethan Carter “NixOS, Nixpkgs, and Nix”. Northeastern University Linux Users Group, Boston, MA, November 19, 2025.
6. Edwards, Ethan Carter “Newline Interactive TVs”. Carroll County Public Schools, Hillsville, VA, December 17, 2024.
7. Edwards, Ethan Carter “Philips Interactive TVs”. Carroll County Public Schools, Hillsville, VA, August 1, 2024.
8. Edwards, Ethan Carter et al. “Cybersecurity Education”. Radford University IMPACT Conference, Abingdon, VA, November 9, 2023.

Journalism Publications, Podcast Appearances

1. Martinez, Tristan; Edwards, Ethan Carter; Mills, Sophia. “[Voter disenfranchisement and political polarization with Ethan Edwards](#)” *HPR Talk*, October 1, 2025.
2. Edwards, Ethan Carter. “[Dual-Party Primaries Don’t Work for Independents](#)” *Harvard Political Review*, September 25, 2025.